Homework 8

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Problem 1

Let

$$A = \begin{pmatrix} -7 & 16 & 8 \\ -4 & 9 & 4 \\ -4 & 7 & 5 \end{pmatrix}.$$

Find out the minimal polynomial of A.

Solution

Calculation gives $\chi_A = (\lambda-1)(\lambda-3)^2$ and $\operatorname{rank}(A-3I) = 2$. The eigenvalue three has geometric multiplicity one and a size-two block. Thus

$$\boxed{m_A(\lambda) = (\lambda-1)(\lambda-3)^2}.$$

Problem 2

Let

$$A = \begin{pmatrix} 0 & 1 & 0 & 0 \\ -1 & 2 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & -4 & -4 \end{pmatrix}.$$

Find out the minimal polynomial of A.

Solution

For the diagonal blocks B, C, $(B-I)^2 = 0$ with $B \neq I$, and $(C+2I)^2 = 0$ with $C \neq -2I$. Thus $m_B = (\lambda-1)^2$, $m_C = (\lambda+2)^2$, and

$$\boxed{m_A = \operatorname{lcm}(m_B, m_C) = (\lambda-1)^2(\lambda+2)^2}.$$

Problem 3

In Problem 3 of Homework 7, A was a matrix whose elementary divisors are

$$\lambda+3, \quad \lambda+3, \quad (\lambda+3)^2, \quad \lambda+2+5j, \quad \lambda+2-5j,$$
$$(\lambda+2+5j)^2, \quad (\lambda+2-5j)^2.$$

Based on that result, find out the minimal polynomial of A.

Solution

Take the largest elementary-divisor exponent at each root, which is two. Combining conjugate factors gives

$$\boxed{m_A(\lambda) = (\lambda+3)^2\big((\lambda+2)^2+25\big)^2}.$$

Problem 4

In Problem 4 of Homework 7, A was a matrix whose elementary divisors are

$$(\lambda+1),\quad (\lambda+1)^2,\quad (\lambda+1+j),\quad (\lambda+1-j),$$
$$(\lambda+1)^2,\quad (\lambda+1+j)^2,\quad (\lambda+1-j)^2.$$

Based on that result, find out the minimal polynomial of A.

Solution

Again the largest exponent at every root is two, so

$$\boxed{m_A(\lambda) = (\lambda+1)^2\big((\lambda+1)^2+1\big)^2}.$$

Problem 5

Let $\mathcal{A} \in L(\mathbb{C}_n[\lambda])$ be defined as

$$\mathcal{A}[f(\lambda)] = f(\lambda+a), \qquad \forall f(\lambda) \in \mathbb{C}_n[\lambda].$$

for a nonzero constant $a \in \mathbb{C}$. Find out

(1) all the generalized eigenvectors of $\mathcal{A}$;

(2) the transition matrix X that reduces A to the Jordan canonical form $J = X^{-1}AX$;

(3) the generating elements $\beta_i(s)$, $i = 1, 2, \ldots, l$, such that

$$\mathbb{C}_n[s] = \mathcal{S}_{\mathcal{A}}[\beta_1(s)] \dotplus \mathcal{S}_{\mathcal{A}}[\beta_2(s)] \dotplus \cdots \dotplus \mathcal{S}_{\mathcal{A}}[\beta_l(s)].$$

Solution

Write the polynomial variable as s, put $\Delta = \mathcal{A} - I$, and use the basis $(1, s, \ldots, s^n)$ for A.

(1) The only eigenvalue is one. For any nonzero f of degree d with leading coefficient c_d,

$$\Delta^d f = c_d d! a^d \neq 0, \qquad \Delta^{d+1} f = 0.$$

Thus every nonzero polynomial is a generalized eigenvector, and a degree-d polynomial has exact order $d+1$.

(2) Let $f_0 = 1$ and take the monomial coefficient columns of f_k as X, with indices starting at zero:

$$f_k(s) = \frac{\prod_{r=0}^{k-1}(s-ra)}{k!a^k}\ (1 \le k \le n), \qquad \boxed{X_{ik} = [s^i]f_k(s)}.$$

Here $[s^i]$ extracts a coefficient. The upper triangular X has nonzero diagonal $1/(k!a^k)$. Since $\mathcal{A}f_0 = f_0$ and $\mathcal{A}f_k = f_k + f_{k-1}$,

$$\boxed{AX = XJ_{n+1}(1), \qquad X^{-1}AX = J_{n+1}(1)}.$$

(3) One generator suffices:

$$\boxed{l = 1, \qquad \beta_1(s) = s^n, \qquad \mathbb{C}_n[s] = \mathcal{S}_{\mathcal{A}}[s^n]}.$$

The polynomials $s^n, \Delta s^n, \ldots, \Delta^n s^n$ have distinct degrees $n, n-1, \ldots, 0$ with nonzero leading coefficients, so they form a basis. Since $\Delta = \mathcal{A} - I$, each lies in $\mathcal{S}_{\mathcal{A}}[s^n]$.